

Dominic Andrew Chomchai

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EDUCATION

University of Wisconsin (Madison, WI)

Sept. 2020 – Present

PhD – Cellular and Molecular Biology | GPA: 3.864/4.0

University of Michigan (Ann Arbor, MI)

Sept. 2016 – May 2020

B.S. – Molecular, Cellular, and Developmental Biology (High Honors) | GPA: 3.664/4.0

- Minor: Spanish

RESEARCH EXPERIENCE

Graduate Research Assistant

Dec. 2020 – Present

University of Wisconsin (Madison, WI) | Research Advisor: William M. Bement

- Thesis title: Exploring the mechanism of positive feedback between Ect2 and Rho in cortical excitability during *Xenopus laevis* cytokinesis.
- Investigated Rho-associated kinase (ROCK) as a negative feedback regulator of Ect2 and RhoA in cortical excitability.
- Collaborated with a biological modeling group and co-published work distinguishing between two mechanistic models of cortical wave propagation.
- Co-developed a Python-based pipeline for batch analysis of excitable and oscillatory dynamics in multichannel time-lapse datasets.
- Developed a Python-based, all-in-one image-processing tool for streamlined image conversion and metadata extraction across commonly used microscopes in the Bement Lab.
- Built custom ImageJ/Fiji batch processing and analysis tools to enhance workflow efficiency and reproducibility in quantitative microscopy experiments.
- Served as the de facto lab manager, coordinating supply orders through communication with lab members, departmental staff, and external vendors.
- Mentored undergraduate researchers in molecular biology and microscopy.

Undergraduate Research Assistant

March 2018 – May 2020

University of Michigan (Ann Arbor) | Research Advisor: Ming Li

- Thesis: The mechanism of the ubiquitination and degradation of lysosomal membrane proteins
- Conducted a whole genome high-throughput screening to identify proteins involved in the assembly of the vacuole membrane protein complex Fth1-Fet5 using fluorescent microscopy in *Saccharomyces cerevisiae*.
- Studied the importance of two distinct lysosomal membrane protein degradation pathways using fluorescent microscopy, ubiquitin blot analysis, and Western blot analysis. Studied the effects of different stress conditions on the general composition of lysosome membrane proteins.
- Conducted a genetic screening of PY motif-containing proteins and analyzed with flow cytometry to identify new adaptors involved in the ubiquitination of lysosomal membrane cargoes.

OUTREACH EXPERIENCE

Special Olympics College at the University of Michigan

September 2016 – May 2020

Vice President (May 2019 – May 2020)

- Led operations in collaboration with the executive board, including running weekly meetings, mentoring new board members, and coordinating cross-university competitions.
- Directed recruitment and outreach efforts by organizing recurring student organization fair booths and engagement events.

Soccer Coach (September 2017 – May 2020)

- Led inclusive weekly practices for athletes with and without intellectual and developmental disabilities, fostering skill development and team cohesion.

WORK EXPERIENCE

USA Hockey, Mount Pleasant, MI and Ann Arbor, MI

September 2010 – March 2020

Referee

- Officiated competitive hockey games across a wide age range, consistently applying rules and maintaining fair, safe play under pressure.
- De-escalated conflicts and communicated rulings clearly and professionally with coaches and players.

LEADERSHIP, TEACHING AND MENTORING EXPERIENCE

Research Mentor, Bement Lab, Madison, WI

Jan. 2023 – Present

- Mentor an undergraduate researcher in molecular biology and microscopy techniques, providing training and ongoing scientific guidance.
- Advise on data interpretation, poster presentations and manuscript-style writing for scholarships.

Teaching Assistant at the University of Wisconsin – Madison, Madison, WI

Biology 570 (Cell Biology)

Aug. 2024 – Dec. 2024

- Developed lesson plans and led optional weekly discussions; designed and administered Canvas quizzes to reinforce lecture and reading concepts.
- Held office hours, graded and provided detailed feedback on 100+ written assignments, and proctored exams.

Biology 151 Lab (Intro Biology Lab Section)

Aug. 2022 – Dec. 2022

- Instructed undergraduates in experimental design and scientific research methods during weekly three-hour lab sections.
- Developed instructional materials, graded lab reports with constructive feedback, and supported students through office hours and exams.

CMB Graduate Program NSF Workshop Leader, Madison, WI

June 2022 – July 2023

- Led workshops mentoring graduate students applying to the NSF GRFP, drawing on personal experience as an Honorable Mention recipient.
- Developed instructional materials and provided detailed feedback on personal and research statements.

After-School Family STEM Program at the Goodman South Library, Madison, WI

Teacher and Co-founder

Sept. 2022 – Nov. 2022

- Co-founded and led a 10-week drop-in family STEM program (K–5 with parents) in partnership with the lead youth librarian, expanding STEM access in an historically underrepresented community.
- Designed adaptable lesson plans and facilitated hands-on experiments across physics, astronomy, and biology to increase family engagement in STEM.

Cellular and Molecular Biology Recruitment Committee

Jan. 2021 and 2022

Virtual Weekend Recruitment Manager

- Coordinated and scheduled virtual recruitment weekends for a graduate training program, organizing interviews between prospective students and current graduate trainees.
- Designed and hosted virtual social events using Gather.town to facilitate engagement and community-building among recruits and students.

The Science Learning Center at the University of Michigan, Ann Arbor, MI *Sept. 2019 – May 2020*

Biochemistry (MCDB 310) Study Group Facilitator

- Led weekly collaborative study sessions, applying evidence-based learning strategies to promote active engagement and mastery of course material.
- Planned structured agendas and guided group discussions to keep sessions focused and productive.

PROFESSIONAL TRAINING

STEM Public Engagement Course (Public Service Fellows), Madison, WI Jan. 2022 – Dec. 2022

- Selected for a competitive, yearlong fellowship focused on effective STEM public engagement, emphasizing asset-based approaches and trust-building with community partners.
- Collaborated weekly with a multidisciplinary graduate cohort to develop strategies for partnering with non-academic organizations and addressing public issues through science.

WIScience Delta Program Research Mentor Training, Madison, WI Sept. 2021 – Dec. 2021

- Completed formal training in effective research mentorship, including evidence-based strategies for communication, expectation-setting, and mentor–mentee problem-solving.

PUBLICATIONS

Chomchai, D., Leda, M., Golding, A., von Dassow, G., Bement, W.M. & Goryachev, A.G. Rho GTPase dynamics distinguish between models of cortical excitability. *Current Biology*, 35(6). doi: 10.1016/j.cub.2025.02.003 (2025).

Yang, X., Reist, L. **Chomchai, D.**, Chen, L., Arines, F. & Li, M. ESCRT, not intraluminal fragments, sorts ubiquitinated vacuole membrane proteins for degradation. *Journal of Cell Biology*, 220(8). doi: 10.1083/jcb.202012104 (2021).

Yang, X., Zhang, W., Wen, X., Bulinski, P., **Chomchai, D.**, Arines, F., Liu, Y., Sprenger, S., Teis, D., Klionsky, D.J. & Li, M. TORC1 regulates vacuole membrane composition through ubiquitin- and ESCRT-dependent microautophagy. *Journal of Cell Biology*, 219(3). doi: 10.1083/jcb.201902127 (2020).

PRESENTATIONS

Seminar Talk. Chomchai, D. & Bement, W. Exploring the mechanics of positive feedback in Ect2-mediated cortical excitability. **Cells, Science and Imaging (CSI): Madison**. Madison, WI, Oct 2025

Subgroup Talk: Frontiers in Cytokinesis (Ten-minute talk). **Chomchai, D. & Bement, W.** Exploring the mechanism of positive feedback in cortical excitability. **The American Society of Cell Biology – Cell Bio 2024**, San Diego, CA, Dec 2024.

Seminar Talk. Chomchai, D. & Bement, W. Positive feedback between Rho and Ect2 is essential for cortical excitability. **Cell Biology Seminar Group**. Madison, WI, March 2024

Subgroup Talk: Frontiers in Cytokinesis (Ten-minute talk). **Chomchai, D. & Bement, W.** Positive feedback between Rho and Ect2 is essential for cortical excitability. **The American Society of Cell Biology – Cell Bio 2022**, Washington D.C., Dec 2022.

POSTERS

Yang, X., Zhang, W., **Chomchai, D.**, Wen, X., Klionsky, D. & Li M. TORC1 regulates the vacuole membrane proteome through ubiquitin- and ESCRT-dependent microautophagy. **MCDB Department Retreat**, Ypsilanti, MI, Dec. 2019.

Chomchai, D. & Bement, W. Positive feedback between Rho and Ect2 is essential for cortical excitability. **The American Society of Cell Biology – Cell Bio 2022**, Washington D.C., Dec 2022.

Chomchai, D. & Bement, W. Positive feedback between Rho and Ect2 is essential for cortical excitability. **The 19th International Xenopus Conference**, Cambridge, MD, August 2023.

Chomchai, D. & Bement, W. Positive feedback between Rho and Ect2 is essential for cortical excitability. **The American Society of Cell Biology – Cell Bio 2023**, Boston, MA, Dec 2023.

Chomchai, D. & Bement, W. Positive feedback between Rho and Ect2 is essential for cortical excitability. **Polarity and Asymmetry in Tissue Architecture and Cell Fate – Gordon Research Conference**, Manchester, NH, June 2024.

Chomchai, D. & Bement, W. Exploring the mechanism of positive feedback in cortical excitability. **The American Society of Cell Biology – Cell Bio 2024**, San Diego, CA, Dec 2024.

Chomchai, D. & Bement, W. Exploring the mechanism of positive feedback in cortical excitability. **The American Society of Cell Biology – Cell Bio 2025**, Philadelphia, PA, Dec 2025.

HONORS AND AWARDS

NSF GFRP Honorable Mention, 2022

Bement Lab, University of Wisconsin-Madison, Madison, WI

- Proposal Title: The Basis of Positive Feedback in Cortical Excitability During Cytokinesis

High Honors for Senior Thesis in Molecular, Cellular, and Developmental Biology, 2022

Ming Li Lab, University of Michigan, Ann Arbor, MI

- Thesis Title: The Mechanism of the Ubiquitination and Degradation of Lysosomal Membrane Proteins

Area 7 Outstanding Volunteer of the Year Award, 2017

Special Olympics Michigan, Mount Pleasant, MI

SKILLS

Laboratory: Molecular cloning design; in vitro RNA synthesis; DNA/RNA purification; PCR and RT-PCR; protein purification; SDS-PAGE and Western blotting; protein degradation assays; siRNA/morpholino knockdown; high-throughput genetic screening; flow cytometry; *Xenopus laevis* fertilization protocols and microinjections; time-lapse fluorescence microscopy (point-scanning, swept-field confocal, widefield, lightsheet); optogenetics; sample fixation and clearing.

Software: Python (NumPy, SciPy, pandas, matplotlib, seaborn, napari, tifffile, pytest); HTML/CSS; JavaScript; ImageJ/Fiji; SnapGene; CellProfiler; PyMOL; GraphPad Prism; Adobe Illustrator/Photoshop; Affinity; PowerPoint/Keynote; Word; Excel; OneNote; EndNote.

Professional: Collaboration; communication; presentation of complex ideas and data; mentoring; academic writing; process optimization; laboratory procurement.